



Stealth fighter

First built for the US Air Force in 1982, but kept secret until 1988, the Lockheed F-117 Nighthawk was the first military aircraft designed around stealth technology. The F-117 was developed after the use of sophisticated surface-to-air missiles in the Vietnam War highlighted the need for reduced radar visibility. After years of development as a top secret “black project,” the F-117 had its maiden flight in Nevada on June 18, 1981. Despite its Fighter “F” designation, the F-117 had no air-to-air combat capabilities—instead, it carried two 2,000-lb (910-kg) laser-guided bombs. It proved a crucial weapon, participating in more than 40 percent of the strategic air strikes in the First Gulf War of 1991.

STEALTH TECHNOLOGY

Despite being almost 66 ft (20.1 m) long, with a wingspan of 43½ ft (13.2 m), the F-117 had the radar signature of a small bird. The craft’s angled surfaces—coated with a matte black, radar-absorbent material (RAM)—scattered incoming radar waves instead of reflecting them back. In order to reduce the craft’s infrared footprint the Nighthawk had no afterburner, and the exhaust was channeled through long, heat-absorbent ducts. Dubbed the “Wobblin’ Goblin” because of its unstable in-flight experience, the F-117 could only be flown using a computer-controlled, fly-by-wire system. Only one F-117 was ever lost in combat—it was shot down in 1999 during the Kosovo War.

Secret but slow, like all attack aircraft the F-117 flew at subsonic speeds because the sonic boom would have revealed its position.